

1 **Fusing short-pulse voltage waveforms with handcrafted features for**
2 **rapid state-of-health estimation of lithium iron phosphate batteries: a**
3 **controlled comparison of encoder pairings**

4 [First Author]^a, [Second Author]^a, [Corresponding Author]^{a,*}

5 ^a [Department, Institution, City Postcode, Country]

6 Corresponding author: [Name].

7 Address: [Postal address].

8 Tel: [].

9 E-mail address: []

10 Declarations of interest: None

11 **Funding information**

12 [Funding sources and grant numbers to be added.]

13 **Abstract:**

14 **Background**

15 Short current pulses offer a rapid diagnostic for the state of health (SOH) of
16 lithium-ion batteries, because a single pulse lasts seconds to minutes and needs only a
17 controlled current step. Whether a learned waveform representation adds information
18 to handcrafted pulse features, and how the encoders of such multi-branch models
19 should be paired, has not been tested under a controlled protocol.

20 **Method**

21 A dual-branch model is built in which a primary branch of handcrafted features
22 and the cumulative cycle count is supplemented by a branch encoding the 101-point
23 pulse-voltage response. All 15 pairings of three handcrafted-feature and five
24 waveform encoders are evaluated on 16,938 pulse records from 64 public lithium iron
25 phosphate (LFP)/graphite cells with five-fold cell-disjoint cross-validation. The model
26 is further tested on an independent nickel–manganese–cobalt (NMC)/graphite dataset,
27 on held-out cycling conditions, under cycle-count errors and sensor noise, and with
28 reduced pulse protocols.

29 **Significant findings**

30 A multilayer perceptron (MLP) handcrafted-feature encoder paired with a patch-
31 based Transformer (PatchTST) waveform encoder attains the lowest cell-level root-
32 mean-square error, 1.377 SOH percentage points (95% CI 1.247–1.513). The
33 waveform branch lowers the error by 8.2% in 58 of 64 cells, most strongly below 80%
34 SOH (17.2%), and the gain is reproduced on the NMC dataset (7.6%, 39 of 44 cells).
35 Encoder rankings depend on the pairing: PatchTST ranks first with the MLP but last
36 with the other two encoders. The model is robust to unseen cycling conditions and
37 $\pm 10\%$ cycle-count errors (at most +3.5%), three charge pulses match the six-pulse
38 protocol, and inference needs only 19,041 parameters. Encoders for pulse-based SOH
39 estimation should therefore be selected as pairings on controlled evidence rather than
40 by architectural complexity.

41 **KEYWORDS** Lithium iron phosphate battery; State of health; Short current pulse;
42 Feature fusion; Patch-based Transformer; Cell-disjoint validation

43 1. Introduction

44 Lithium-ion batteries (LIBs) in electric vehicles and energy storage systems age
45 throughout their service life. Assessing their state of health (SOH) is essential for
46 estimating available energy and power, planning maintenance and informing
47 decisions on second-life use [1–3]. Capacity-based SOH is defined as the ratio of a
48 battery’s current capacity to its initial capacity, and its direct measurement requires a
49 full charge–discharge test under controlled conditions. Such tests typically take
50 several hours and interrupt normal operation, making frequent field testing impractical
51 [4,5]. SOH must therefore be inferred from signals that can be acquired more rapidly.
52 Data-driven approaches have been built on differential curves, partial charging curves,
53 voltage relaxation, electrochemical impedance spectroscopy (EIS) and short current
54 pulses. Most of them impose demanding operating requirements, such as prolonged

55 data collection, complete controlled cycles or dedicated excitation hardware [1,4,6,8–
56 11]; differential-curve peaks, for example, are further blurred by sensor noise and by
57 the flat voltage plateau of lithium iron phosphate (LFP) cells [7]. These requirements
58 are difficult to meet outside the laboratory. Short current pulses are a notable
59 exception. A current pulse is essentially a time-domain impedance measurement [12],
60 and under suitable test conditions its response can even be used to reconstruct
61 impedance spectra [13]. Degradation modes such as loss of lithium inventory, loss of
62 active material and interfacial film growth alter the pulse response, causing it to
63 evolve with ageing [12,14], and pulse-based characterisation is already standard in
64 battery testing [15]. Whereas a full hybrid pulse power characterisation sequence can
65 exceed 12 h [17], a single pulse lasts only seconds to minutes and requires nothing
66 more than a controlled current step, making pulse testing a rapid and practical route to
67 capacity estimation [16]. For LFP cells, the pulse response is dominated by kinetic
68 overpotentials, so its change with ageing remains measurable even where the
69 equilibrium voltage barely moves.

70 Data-driven SOH estimation relies on input features extracted from voltage,
71 current, temperature or impedance measurements, and studies adopt diverse forms—
72 voltage curves, incremental-capacity or differential-voltage curves, statistical metrics,
73 frequency-domain indicators or model parameters—depending on battery type,
74 operating condition and model structure. Deployment in battery management systems
75 (BMSs) adds further constraints: inference must run in real time on resource-limited
76 hardware, and black-box behaviour undermines trust in safety-relevant decisions
77 [18,19]. By how they are obtained, the inputs fall into two categories [21,22]. One
78 extracts handcrafted health indicators from voltage, current, temperature or
79 incremental-capacity curves as inputs to regression models [3]. These indicators are

80 physically interpretable but rely on human expertise and transfer poorly across
81 operating conditions [21,22]. The other learns representations directly from raw
82 sequences with deep networks; it avoids manual feature design but offers limited
83 interpretability, and its performance depends heavily on how well the network
84 architecture matches the task [18,22]. Multi-branch models can combine the strengths
85 of both by fusing several input representations [23–25]; however, their encoders are
86 selected at design time, and comparative evaluations focus on baseline models and
87 fusion strategies rather than on systematic comparisons of branch encoders [24,25].
88 Systematic comparisons under consistent protocols and rigorous cross-validation also
89 remain limited [26], and some benchmarks consider only a single input type [27].
90 Consequently, there is no consensus on feature extraction, and this fragmentation,
91 together with the lack of standard evaluation protocols, makes it difficult to compare
92 methods on a common basis and to transfer them to new batteries or real BMSs [20].

93 For SOH estimation from short current pulses, existing studies each use a single
94 input representation: Li et al. [16] and Nowacki et al. [28] use raw pulse responses,
95 whereas Tao et al. [17,29] and Ran et al. [30] regress SOH on features extracted from
96 the responses, and transfer across states of charge has also been examined [31]. These
97 studies confirm that short pulses carry SOH-relevant information, but no consensus
98 has been reached on how features should be extracted or which representation is
99 preferable. Three issues remain unclear. First, the main effect of the encoder in each
100 branch is unknown, including whether a more complex encoder for handcrafted
101 features improves performance. Second, encoder pairing has not been examined: the
102 performance of a waveform encoder may depend on the handcrafted-feature encoder
103 it is paired with, a dependence that comparisons varying only one branch cannot
104 capture. Third, comparability across studies is limited by differences in input

105 information [32–34], data splitting and preprocessing [35,36] and training settings
106 [26], and no common benchmark exists [37]. A systematic comparison is therefore
107 needed in which both encoders are varied and all combinations are evaluated under a
108 unified protocol. The answer also matters for deployment: if more complex encoders
109 bring no benefit, lightweight structures should be preferred for resource-limited
110 BMSs.

111 Fusing handcrafted features with raw waveforms can improve SOH estimation
112 [23–25,42]. However, previous studies fused signals from different measurements
113 [42,43] or different transforms of one signal [24,25], and few have addressed the
114 fusion of handcrafted features and raw waveforms derived from the same pulse; the
115 design of a supplementary encoder for a single short pulse has received little attention.
116 Handcrafted features as the primary branch are common practice in SOH estimation
117 [3,17,29,30]. They have clear physical meanings and suit lightweight models [21], in
118 line with BMS requirements for interpretability and cost, and context absent from the
119 raw waveform, such as usage history, can be added to them explicitly. However,
120 handcrafted features compress the response: each describes only a predefined
121 property, so coverage of all ageing-related changes across operating conditions cannot
122 be assured [22]. Moreover, most handcrafted features are differences, statistics or
123 frequency-domain quantities computed individually, so joint changes across different
124 parts of the transient are difficult to characterise. A learned encoder can extract such
125 shape information directly from the raw waveform, including information not covered
126 by handcrafted features. Used alone, however, learned encoders offer limited
127 interpretability [18], and raw waveforms carry no information on operating conditions
128 or usage history. A learned waveform encoder is therefore better suited to
129 supplementing handcrafted features than to replacing them.

For the waveform branch, patch-based encoders are a promising candidate. PatchTST encodes contiguous segments as tokens and relates them through self-attention [40], which may complement individually computed features. However, it was designed for long-sequence forecasting [40]. In battery research it has been applied to SOH estimation [44], lifetime prediction [45] and voltage prediction [46], but its adaptation to short-pulse inputs and scalar health estimation remains unexamined.

To address these issues, we construct a complete comparison matrix of waveform and handcrafted-feature encoders under a unified experimental protocol, with identical data splits, preprocessing and training budgets. The matrix includes five waveform encoders with convolutional, fully connected, temporal convolutional [38], point-wise attention [39] and patch-based [40] architectures, and three handcrafted-feature encoders: a multilayer perceptron (MLP), a ResNet and an FT-Transformer [41]. All combinations are compared using cell-disjoint cross-validation. Unlike fusion studies that select branch encoders in advance [23–25] or model comparisons using a single input [27], this design estimates main effects, tests dependence on encoder pairing and attributes error differences to the encoders themselves. Based on the matrix, we develop a dual-branch model consisting of a primary handcrafted-feature branch, encoded by an MLP and incorporating operating context and cumulative cycling information, and a supplementary branch of patch-based waveform representations, for which a concrete adaptation of PatchTST from time-series forecasting to scalar SOH estimation is presented. The two branches are fused at a late stage. To reflect the general decline of capacity with cycling, a soft monotonicity constraint on predicted SOH with respect to the cumulative cycle count is imposed; unlike approaches built on degradation dynamics, mechanistic features or

digital twins [47–49], it requires no mechanistic model. Ablation studies and cell-level statistical analysis quantify the contribution of each design factor.

The main contributions of this work are as follows:

- (1) We construct a complete comparison matrix of waveform and handcrafted-feature encoders under a unified experimental protocol, enabling comparable evaluation of their main effects and of their dependence on encoder pairing with identical data splits, preprocessing and training budgets.
- (2) We develop a dual-branch model for SOH estimation from short current pulses, using a primary handcrafted-feature branch supplemented by patch-based waveform representations, with operating context, cumulative cycling information and a concrete adaptation of PatchTST to scalar SOH estimation. Ablation studies and cell-level statistical analyses isolate the contributions of individual design factors and identify the conditions under which the waveform branch helps.
- (3) We examine the generality of the model on held-out cycling conditions, under cycle-count errors and sensor noise, with reduced pulse protocols and on an independent nickel–manganese–cobalt (NMC)/graphite dataset, on which the waveform increment is reproduced, and quantify its attribution and computational cost.

2. Data and unified protocol

All models share one data protocol, so that error differences between encoders can be attributed to architecture: the cleaning rules, cell split, input policy of both branches and the train-fold feature selection and scaling are identical across the 15 encoder pairings and all supplementary experiments (Table 1).

2.1. Dataset, pulse protocol and SOH definition

The public UConn-ISU-ILCC LFP/Gr battery ageing dataset [28] contains 64 LFP/graphite power cells (nominal capacity 1.2 Ah) aged under 11 cycling conditions that differ in constant-current–constant-voltage (CC–CV) charge rate (2C–3.25C), constant-current discharge rate (1.5C or 2C) and depth of discharge (DOD, 60–100%) (Supplementary Table S1). Cycling was interrupted roughly every 100 cycles for a reference performance test (RPT) that measured the remaining capacity and applied short pulses [28,31]. All ageing tests and RPTs were performed by the data publisher; this study reanalyses the data.

Each RPT applies one charge and one discharge pulse at each of three SOC. A pulse consists of 30 s at 0.2C, 10 s at 1C and a 60 s rest (Fig. 1a) [31]; it lasts 100 s and passes about 0.44% of the nominal capacity. Pulses are triggered at preset voltage cut-offs, so 20%, 50% and 90% are nominal SOC at beginning of life and the actual SOC drifts as capacity fades [31]. Each RPT therefore yields records for six conditions (three nominal SOC $\times$ two directions). Within the 1 Hz record, the voltage step at each current change mainly reflects ohmic resistance and fast charge transfer, the continued change during constant current adds diffusion, and the slow recovery during rest is dominated by diffusion relaxation [12] (Fig. 1b,c). The voltage difference between the ends of the 0.2C and 1C segments gives the 10-s direct-current internal resistance (DCIR) used in Section 2.3.2.

SOH is normalised by the discharge capacity at the first RPT of the same cell:

$$y_{ir} = 100 \times \frac{Q_{ir}}{Q_{i0}} \quad (1)$$

where Q_{ir} is the discharge capacity of cell i at RPT r . SOH thus denotes capacity retention relative to the initial measurement rather than to the rated capacity. The capacity comes from the full reference discharge of the RPT, i.e. the hour-long test

that pulse diagnostics aim to replace; it serves only as the label, and no quantity derived from it enters the model (Section 2.3.2).

Each record contains 101 voltage samples: sample 0 is the last voltage before current is applied, and the 30, 10 and 60 s segments were resampled to 1 Hz by piecewise cubic Hermite interpolation in the publisher's processing code [31]. Interpolated points are not treated as independent samples. Each inference uses one pulse record, i.e. one 100-s test. The primary metric averages the six single-record predictions of an RPT, which corresponds to a joint diagnosis at three nominal SOC's whose duration also includes bringing the cell to each SOC; single-condition errors are reported separately (Section 4.3).

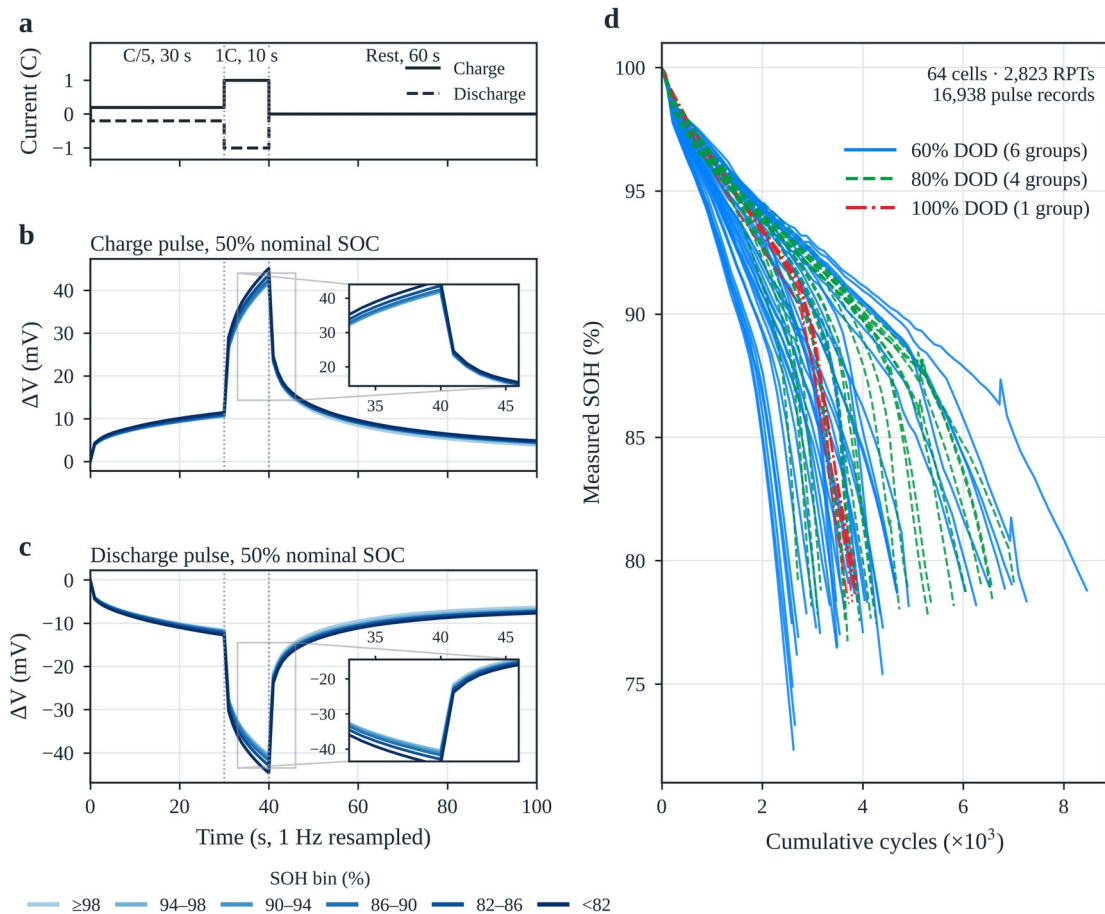


Fig. 1. Dataset and pulse response. (a) Programmed current of the charge (solid) and discharge (dashed) pulses: 30 s at C/5, 10 s at 1C and 60 s rest. (b, c) Mean voltage response ΔV relative to the pre-pulse voltage at 50% nominal SOC, grouped by measured SOH (darker

218 = lower SOH); insets magnify the end of the 1C segment and the onset of relaxation, where
 219 the overpotential grows with ageing. (d) Measured SOH of the 64 cells versus cumulative
 220 cycle count, coloured by depth of discharge; lines connect measured RPTs without
 221 interpolation.

222 2.2. Data cleaning and cell-disjoint split

223 Before cleaning, the data contain 17,130 records in 2,855 cell–RPT groups. A
 224 record is retained only if it passes all predefined checks:

$$225 \quad A = \left\{ (V, x, y) : \begin{array}{l} V_t \in [2.5, 4.5] V \ \forall t; \max_t V_t - \min_t V_t \in [0.005, 0.5] V; \max_t |V_{t+1} - V_t| \leq 0.1 V \\ y \in [0, 110]; q_{miss} \leq 0.20; |G_{i,r}| = 6; \max_{g \in G_{i,r}} y_g - \min_{g \in G_{i,r}} y_g \leq 10^{-5} \end{array} \right\} \quad (2)$$

226 where V_t is the t -th voltage sample, y the SOH in percent, q_{miss} the fraction of
 228 missing handcrafted features and $G_{i,r}$ the six condition records of cell i at RPT r . All
 229 192 removed records failed because their cycle count was not finite: the publisher’s
 230 data lack the cycle count at the 42nd RPT, which affects all 32 cells that reached it (32
 231 complete cell–RPT groups). The remaining 16,938 records form 2,823 complete
 232 groups covering all 64 cells, with SOH from 72.34% to 100% (Fig. 1d). Thresholds
 233 were fixed in advance and logged, not tuned on test errors.
 234

235 Five outer test folds were built with split seed 42: the 64 cell identifiers were
 236 permuted and divided into five disjoint test sets. Within the non-test cells of each fold,
 237 validation cells were drawn with the fold index as offset and a stride of five, and the
 238 remaining cells were used for training, giving 40/11/13, 41/10/13, 41/10/13, 41/10/13
 239 and 42/10/12 training/validation/test cells. All RPTs and conditions of a cell are
 240 assigned together, because random record-level splits inflate accuracy [36]. The split
 241 is identical for all experiments, and the MLP backbone with the PatchTST
 242 complement serves as the reference configuration R0.

Table 1. Data, inputs and experimental design.

Item	Setting
Data	Public UConn-ISU-ILCC LFP/Gr dataset; 64 cells, 16,938 records, 2,823 cell-RPT groups
Pulse conditions	Nominal SOC 20%, 50%, 90% $\times$ charge, discharge; 100 s per pulse
Waveform branch	101-point ΔV , per-position train-fold standardisation
Backbone branch	50 train-fold-selected features (49 waveform-derived + DCIR) + cumulative cycle count τ
Excluded inputs	Coulomb-counted SOC and its deviation (derived from reference capacity); other records of the same RPT; past capacities; cell identity
Validation	Five cell-disjoint outer folds; initialisation seeds 0, 1, 2
Encoder matrix	5 waveform $\times$ 3 backbone encoders $\times$ 5 folds $\times$ 3 seeds = 225 trainings
Further experiments	19 neural configurations $\times$ 15 = 285 trainings; 30 reproduced reference models; conventional regressors (Section 3.5)
Primary metric	Cell-macro RMSE after averaging the six condition predictions of each RPT

2.3. Handcrafted backbone inputs

2.3.1. Candidate feature families

The handcrafted branch is the backbone of the model. Of 150 candidate features, 143 are computed from the voltage of a single pulse and 7 describe condition and age (Table 2). Waveform-derived features follow the publisher’s pulse timing and act on the 30-s low-current segment, the 10-s high-current segment, the 60-s rest and the full sequence. They form six families, each feature having a fixed window and formula (Supplementary Note S1). All are computed item by item: transient-boundary features span adjacent segments but compare only predefined sample pairs and do not represent joint shape changes across the transient, which motivates the waveform complement (Section 2.4).

Table 2. Candidate handcrafted feature families and their selection on the training cells of fold 0.

Family	Candidates	Window	Content	Selected
Segment statistics	45	30 s, 10 s, 60 s segments	Level, dispersion, shape, entropy, linear trend	9
Global waveform	24	Full sequence	Global statistics, end points, quantiles	7
Difference dynamics	22	First and second differences	Rate of voltage change and its change	14

Family	Candidates	Window	Content	Selected
Transient boundary	18	Predefined sample pairs	Steps at current changes, net changes, response times	2
Frequency and wavelet	24	Demeaned, windowed sequence	Spectral amplitudes, band energies, spectral shape, wavelet energies	16
Nonlinear complexity	10	Full sequence	Autocorrelation, Hjorth parameters, entropies, Hurst exponent	1
Condition and age context	7	Record metadata	Nominal and coulomb-counted SOC, deviation, direction, DCIR, cycle count, RPT index	1 ^a
Total	150			50

^a Matched-condition DCIR. Across the five folds, 47 features were selected in every fold and 54 in at least one. Coulomb-counted SOC, SOC deviation, cumulative cycle count and RPT index do not enter selection (Section 2.3.2).

2.3.2. Condition context and cumulative cycle count

A single pulse carries no record of cumulative use, and its starting SOC drifts with ageing. The backbone therefore considers seven context candidates,

$$\left[\frac{SOC_{nom}}{100}, \frac{SOC_{coul}}{100}, \frac{SOC_{coul} - SOC_{nom}}{100}, \pm 1, DCIR_{pulse, SOC}, n_{i,r}, r \right] \quad (3)$$

namely nominal SOC, coulomb-counted SOC, their deviation, pulse direction, matched-condition DCIR, cumulative cycle count and RPT index. The publisher computes the DCIR from the voltage difference between the two current levels of the same pulse, $R = |V_{40} - V_{30}| / (1.2 A - 0.24 A)$ [31], where V_{30} and V_{40} are the last samples of the 0.2C and 1C segments. It depends only on the pulse, represents the power-related resistance discussed in Section 1, and enters feature selection.

Coulomb-counted SOC and its deviation from the nominal SOC are excluded. The publisher's code divides the charge accumulated at the pulse start by the capacity measured in the reference discharge of the same RPT [31], which is the numerator of the SOH label in Eq. (1); both quantities are therefore label-derived and unavailable without a full capacity test. Nominal SOC and pulse direction remain candidates, but since every RPT contains all six conditions their rank correlation with SOH is zero

276 and they were never selected; the direction is also evident from the sign of the
277 waveform input.

278 The cumulative cycle count does not enter selection; it is standardised separately
279 as the input τ (Section 2.5), and the RPT index is removed to avoid duplicating age
280 information. The count $n_{i,r}$ is accumulated from the cycle intervals between
281 successive RPTs. Because the 42nd-RPT interval is missing, the 32 affected cells
282 undercount one interval (67–118 cycles, median 85, i.e. 1.3–2.3% of life) at their
283 subsequent cell–RPT points; Fig. 1d and the model use the same rule. Using τ
284 requires the cell’s cycle record, which defines the application scope of the model.

285 2.3.3. Train-fold feature selection

286 Removing cumulative cycle count, RPT index and the two label-derived SOC
287 quantities leaves 146 selectable features. In each fold, the Spearman rank correlation
288 between each feature and SOH is computed on the training records only,

$$289 \quad \rho_j = \frac{\sum_i (r_{ij} - \bar{r}_j)(r_i^y - \bar{r}^y)}{\sqrt{\sum_i (r_{ij} - \bar{r}_j)^2 \sum_i (r_i^y - \bar{r}^y)^2}} \quad (4)$$

290 where r_{ij} and r_i^y are the ranks of feature j and of the label. Features are taken
291 greedily in descending $|\rho_j|$, skipping any candidate whose absolute rank correlation
292 with an already selected feature reaches 0.98:

$$293 \quad S \leftarrow S \cup \{j\} \text{ if and only if } \max_{i \in S} |\rho_{ji}^{rank}| < 0.98 \text{ and } |S| < K \quad (5)$$

294 If fewer than K features pass, the remaining finite-correlation features are added
295 in descending $|\rho_j|$ without the redundancy constraint. Each fold selected 50 features,
296 47 of them in all five folds (54 in the union), so the selection is stable across training
297 cells. In fold 0, frequency and wavelet (16) and difference-dynamics features (14)
298 dominate (Table 2), and the DCIR ranks 13th–14th in every fold. The dominant
299 frequency and the 85% spectral roll-off are constant within the training cells of some

300 folds and are skipped, so at most 144 features can be ranked. The choice $K=50$ is
 301 justified in Section 4.5.3.

302 2.4. Waveform complement input

303 The waveform branch receives only the voltage sequence of the pulse, without
 304 condition or age information. The sequence $V_0, \dots, V_{100}$ is referenced to its first
 305 sample,

$$306 \quad \Delta v_t = V_t - V_0, t = 0, 1, \dots, 100 \quad (6)$$

307 which removes the starting voltage so that the sequence represents the pulse-
 308 induced polarisation, the kinetically controlled part that remains measurable on the
 309 flat LFP plateau. Each position is then standardised with training-fold statistics,

$$310 \quad \tilde{v}_t = \frac{\Delta v_t - \mu_t^w}{\sigma_t^w} \quad (7)$$

311 so the full 101-point transient is retained for the waveform encoder to learn inter-
 312 segment shape information not covered by the handcrafted features.

313 2.5. Imputation, scaling and input audit

314 Missing values are imputed by the training-fold median of finite values and then
 315 standardised:

$$316 \quad m_j = \text{median}\{x_{ij} : x_{ij} \in R\}, \tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \in R \\ m_j, & x_{ij} \notin R \end{cases}, z_{ij} = \frac{\tilde{x}_{ij} - \mu_j}{\sigma_j} \quad (8)$$

317 The cumulative cycle count is standardised and appended to the K selected
 318 features,

$$319 \quad n_{i,r} = \sum_{q \leq r} \Delta c_{i,q}, \tau = \frac{n_{i,r} - \mu_n}{\sigma_n}, h = [z_1, \dots, z_K, \tau]^T \in R^{K+1} \quad (9)$$

320 giving a 51-dimensional backbone input for $K=50$. Labels are divided by 100
 321 and standardised, and predictions are transformed back:

$$322 \quad y^* = \frac{y/100 - \mu_y}{\sigma_y}, \hat{y} = 100(\sigma_y \hat{y}^* + \mu_y) \quad (10)$$

All statistics in Eqs. (4), (5) and (7)–(10) are fitted on the training fold only. Apart from the cycle count, every input is obtained from the single pulse; other condition records of the same RPT, past capacities, cell identity and reference-capacity-derived quantities never enter the model (Supplementary Table S2).

3. Model and experimental design

3.1. Framework

The model consists of a handcrafted backbone encoder E_h , a waveform complement encoder E_v and a regression head g (Fig. 2). The backbone receives the 51-dimensional input h of Eq. (9) and the complement receives $\tilde{v}$ of Eq. (7); both are derived from the same pulse record, and τ additionally needs the cycle record. Each encoder outputs a 32-dimensional embedding, and the concatenation is mapped to the standardised SOH:

$$\hat{y}^* = g\left([E_v(\tilde{v}); E_h(h)]\right) \quad (11)$$

The two branches differ in information: the backbone carries pulse-derived physical descriptors such as the DCIR together with usage information and can estimate SOH on its own, whereas the complement sees only the shape of the response. The complement is therefore evaluated by its increment over the backbone (Section 4.2). In the encoder matrix, three backbone and five complement candidates are crossed while everything else is held fixed.

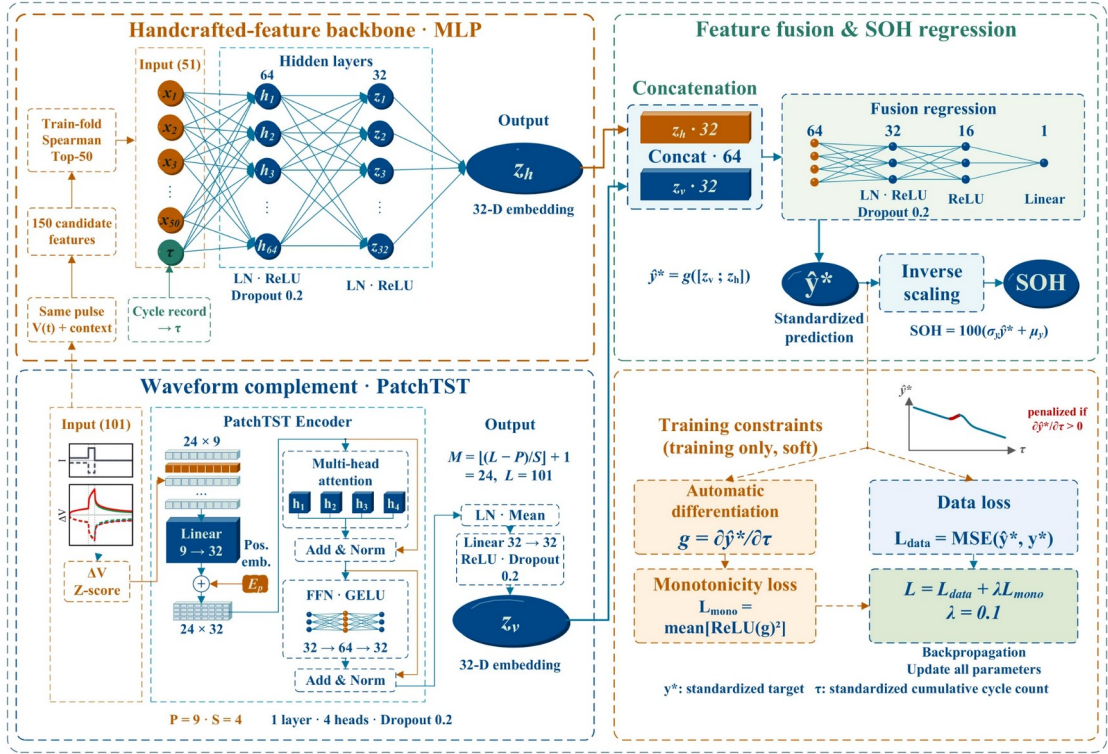


Fig. 2. Dual-branch SOH estimation model (configuration R0). Top left: handcrafted backbone; 50 train-fold-selected features and the standardised cumulative cycle count τ form a 51-dimensional input encoded by an MLP into a 32-dimensional embedding. Bottom left: waveform complement; the 101-point pulse response is referenced to its first sample, standardised per position and split into 24 patches (length $P = 9$, stride $S = 4$) that a single-layer PatchTST encodes into a 32-dimensional embedding. Right: the concatenated 64-dimensional embedding is regressed to the standardised SOH $\hat{y}^*$ and inverse-transformed. Training minimises the mean squared error plus a penalty on positive $\partial \hat{y}^* / \partial \tau$ ($\lambda = 0.1$); the soft constraint acts only during training.

3.2. Backbone encoders

Three backbone encoders are compared. The MLP maps the 51 inputs to 64 and then 32 units, each layer followed by layer normalisation and ReLU, with dropout 0.2 after the first layer (5,600 parameters). ResNet and FT-Transformer follow their tabular reference implementations [41], adjusted to a 32-dimensional output at a similar budget: ResNet uses two residual blocks of width 28 (5,800 parameters), and FT-Transformer tokenises each input to width 20 and uses one Transformer block with four heads and a ReGLU feed-forward layer of hidden width 26 (6,124 parameters), both with dropout 0.2. All three receive the same input; τ is passed as a

361 separate tensor and concatenated so that the partial derivative of the prediction with
 362 respect to τ can be computed (Section 3.4).

363 3.3. Waveform encoders and PatchTST adaptation

364 Five complement encoders map the 101-point standardised sequence to a 32-
 365 dimensional embedding, each ending with a linear projection, ReLU and dropout 0.2
 366 (Table 3). The convolutional neural network (CNN) has three convolutional layers
 367 (channels $1 \rightarrow 16 \rightarrow 32 \rightarrow 64$, kernels 7, 5, 3, each with batch normalisation and ReLU,
 368 max pooling after the first two) followed by global average pooling; the MLP maps
 369 $101 \rightarrow 64 \rightarrow 48 \rightarrow 32$; the temporal convolutional network (TCN) [38] has width 24 and
 370 three residual blocks with dilations 1, 2 and 4, using symmetric padding over the
 371 complete pulse rather than strictly causal convolution; and the point-wise Transformer
 372 (PW-Trans.) [39] treats each of the 101 samples as a token (width 24, one layer, four
 373 heads) with mean pooling over tokens.

374 PatchTST was introduced for long-horizon forecasting [40]; here its patching
 375 idea is retained. The sequence of length $L=101$ is split into overlapping patches of
 376 length P and stride S , whose number is

$$377 \quad M = \left\lfloor \frac{L-P}{S} \right\rfloor + 1, p_m = (\tilde{v}_{(m-1)S}, \tilde{v}_{(m-1)S+1}, \dots, \tilde{v}_{(m-1)S+P-1})^T \in \mathbb{R}^P \quad (12)$$

378 With $P=9$ and $S=4$, $M=24$ and the last patch ends exactly at the 101st sample,
 379 so no padding or truncation is needed. Each patch is linearly embedded to $D=32$, a
 380 learnable positional embedding is added, and a single post-normalisation Transformer
 381 encoder layer with four heads, a GELU feed-forward width of 64 and dropout 0.2
 382 models patch-level dependencies; layer normalisation, mean pooling over tokens and
 383 a linear projection give the 32-dimensional embedding (Supplementary Note S2).
 384 Patching first represents neighbouring samples jointly within a local window and then

relates segments through attention, reducing the token count from 101 to 24; the branch has 10,752 parameters.

Relative to the original PatchTST [40], four changes adapt it to single-value regression from one short pulse. (i) Mean pooling plus a linear projection replaces the flatten head, making the embedding size independent of the patch count; a flatten head would raise the branch to 34,304 parameters (42,593 in total). (ii) Reversible instance normalisation (RevIN) is not used, preserving the response amplitude, which scales with internal resistance and may carry ageing information. (iii) The input is the first-sample-referenced, position-standardised single channel of Section 2.4 rather than the raw voltage. (iv) Only one encoder layer is kept, without the multi-step forecasting head or self-supervised pre-training. Changes (i)–(iii) are tested by reverting each in turn (Section 4.5.2).

Table 3. Architectures and parameter counts of the backbone and complement encoders.

Module	Encoder	Main settings	Parameters
Backbone	MLP	51 → 64 → 32; layer norm, ReLU; dropout 0.2 after layer 1	5,600
Backbone	ResNet	2 residual blocks; width 28; hidden width 28	5,800
Backbone	FT-Transformer	1 block; token width 20; 4 heads; ReGLU width 26	6,124
Complement	CNN	Channels 1 → 16 → 32 → 64; kernels 7, 5, 3; pooled projection	11,232
Complement	MLP	101 → 64 → 48 → 32; layer norm, ReLU, dropout	11,440
Complement	TCN	Width 24; 3 residual blocks; dilations 1, 2, 4; symmetric padding	11,744
Complement	PW-Trans.	101 tokens; width 24; 4 heads; 1 layer; FFN 48	8,192
Complement	PatchTST	24 patches; width 32; 4 heads; 1 layer; FFN 64; mean-pooled projection	10,752
Head	MLP	Concatenated 2×32 ; 64 → 32 → 16 → 1	2,689

Total parameters of a pairing are the sum of backbone, complement and head: 16,481–20,557 across the 15 pairings and 19,041 for R0.

3.4. Regression head, loss and training

The head maps the 64-dimensional concatenation through $64 \rightarrow 32 \rightarrow 16 \rightarrow 1$ fully connected layers, with layer normalisation, ReLU and dropout 0.2 after the first layer and ReLU after the second (2,689 parameters). R0 thus has $5,600 + 10,752 + 2,689 =$

19,041 parameters, and the similar budgets of all 15 pairings (16,481–20,557) limit the influence of model size on the comparison. Linear weights use Xavier uniform initialisation with zero biases.

The training objective combines the mean squared error of the standardised SOH with a penalty on positive partial derivatives with respect to τ :

$$L = M S E(\hat{y}^*, y^*) + \frac{\lambda}{B} \sum_{b=1}^B [\partial \hat{y}_b^* / \partial \tau_b]_+^2 \quad (13)$$

where B is the mini-batch size, $[x]_+ = \max(0, x)$ and $\lambda = 0.1$. The term discourages predictions that rise with τ when other inputs are held fixed, encoding the prior that capacity fades with cycling without a mechanistic degradation model. Because the voltage response and handcrafted features also change with ageing, a non-positive partial derivative does not guarantee monotone predictions along a measured trajectory, so the penalty is a soft constraint whose trajectory-level effect is examined in Section 4.5.1.

All networks use AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch size 256, at most 100 epochs and gradient-norm clipping at 5. ReduceLROnPlateau (patience 5, factor 0.5), early stopping with patience 15 and best-epoch selection all monitor the cell-macro root-mean-square error (RMSE) on the validation cells. Each configuration is trained on the five outer folds with initialisation seeds 0, 1 and 2; mini-batch shuffling uses a separate generator (seed + 10,000) and deterministic CUDA kernels, so every run is bitwise reproducible. Training used an NVIDIA GeForce RTX 5060 Laptop GPU; the full procedure is given in Supplementary Algorithm S1.

3.5. Ablations and reference models

3.5.1. Ablation and sensitivity configurations of $R0$

Branch ablations test the necessity of each branch (Supplementary Table S3). A2 keeps only the backbone (with τ and the monotonicity penalty) and feeds its 32-

429 dimensional embedding to a $32 \rightarrow 32 \rightarrow 16 \rightarrow 1$ head (7,265 parameters); it is the direct
430 reference for the waveform increment. A1 keeps only the PatchTST branch and
431 concatenates the DCIR and τ directly to a $34 \rightarrow 32 \rightarrow 16 \rightarrow 1$ head (12,481 parameters).
432 Factor ablations remove the penalty ($\lambda=0$, A3) and additionally remove τ (A4, 50-
433 dimensional backbone input). Patch parameters are compared on a 2×2 grid: P1 (5,
434 2), P2 (5, 4), P3 (9, 2) and R0 (9, 4). Adaptation ablations revert one change of
435 Section 3.3 at a time: D1 uses a flatten head, D2 adds RevIN with learnable affine
436 parameters before patching, and D3 uses the raw voltage instead of Δv (keeping
437 position-wise standardisation). Finally, K is varied over {10, 20, 30, 50, 75, 100,
438 144}, where 144 is the number of rankable candidates.

439 3.5.2. *Conventional regressors and information-source controls*

440 To test whether the backbone must be neural, five conventional regressors are
441 trained on exactly the 51-dimensional input of A2: ridge regression, random forest, a
442 histogram gradient-boosting decision tree (GBDT; scikit-learn, the same algorithm
443 family as LightGBM), support vector regression with a radial basis function kernel
444 (RBF-SVR) and Gaussian process regression (GPR). Hyperparameters are chosen on
445 the validation cells from predefined grids (Supplementary Note S3) and are not
446 refitted on training plus validation cells. To separate information sources, ridge,
447 random forest and GBDT are also trained on τ alone and on the 50 features without τ .

448 3.5.3. *Reproduction of a published pulse model*

449 The raw-pulse model of Nowacki et al. [28] is reproduced with the defaults of
450 the public code under the present split and metric: the input is the 101-point Δv ,
451 inputs and targets are standardised with training statistics, and an MLP with five
452 hidden layers of 100 tanh units is trained with Adam (learning rate 0.0015, batch 50,
453 at most 500 epochs, early stopping on training loss with patience 5). One model per

454 pulse direction covers all SOCs, and only the SOH output of the original multi-output
455 model is kept (30 models in total).

456 3.5.4. *Waveform increment over a strong backbone*

457 Because the increment may depend on backbone strength, the GBDT selected in
458 Section 3.5.2 is used as a backbone and networks learn its residuals. Training-cell
459 residuals come from four-fold cell-disjoint out-of-fold predictions, so that they are not
460 shrunk by in-sample fitting; validation and test residuals use a GBDT fitted on all
461 training cells. The residual learners G+R0, G+A2 and G+A1 share the structures and
462 training of R0, A2 and A1, with the residual standardised as the target, and the final
463 prediction is the sum of the GBDT and residual predictions. (G+R0) – (G+A2) is thus
464 the waveform increment over the GBDT backbone. As a post hoc addition, the 101
465 standardised voltages were also appended directly to the GBDT and SVR inputs and
466 re-tuned.

467 3.5.5. *Generalisation, robustness and deployment experiments*

468 Five further experiments probe the scope of the model. (i) Unseen cycling
469 conditions: each of the 11 cycling groups is held out in turn (leave-one-group-out),
470 with validation cells drawn from the remaining groups. (ii) Independent chemistry:
471 the UConn-ILCC NMC/Gr dataset [31] (44 cells, 11 cycling conditions) uses the same
472 100-s pulse protocol; the three nominal SOCs 20%, 50% and 90% are used for parity,
473 giving 8,016 records in 1,336 cell-RPT groups (SOH 41.2–100%), and the identical
474 pipeline of the original implementation is applied with five cell-disjoint folds (seed
475 42); because the DCIR enters none of the 50 selected features on these data, it is
476 added to the input of A1 separately, as its definition requires. Only the voltage-step
477 and range limits of Eq. (2) are relaxed to 0.3 V and 0.8 V for the higher-resistance
478 NMC cells, which removes no record. (iii) Robustness: the cycle count of the test
479 cells is biased by $\pm 10\%$ with the trained models fixed, and zero-mean Gaussian noise

with $\sigma_{\text{noise}} = 1$ or 2 mV is added to all voltages before feature extraction, with every model retrained on the noisy data (matched sensor noise). (iv) The number of pulses averaged per RPT is varied from one to six over all condition combinations. (v) Integrated gradients [50] attribute R0's prediction to the waveform samples, and parameters, floating-point operations (FLOPs) and single-thread CPU latency quantify deployment cost. Except for (ii), these experiments were run with an independent CPU re-implementation of the protocol. Under the original five-fold split, the re-implementation gives RMSEs of 1.427, 1.474 and 1.028 pp for R0, A2 and GBDT (versus 1.377, 1.500 and 1.022 pp), and every comparison in Section 4.6 is made within the re-implementation (Supplementary Note S6).

3.6. Metrics and statistical tests

The six condition records of an RPT share one capacity label and cannot be treated as independent samples. For cell i , RPT r and initialisation s , the six condition predictions are first averaged to $\bar{y}_{irs}$; the RMSE of each cell over its RPTs is then averaged over seeds and cells:

$$e_{is} = \sqrt{\frac{1}{R_i} \sum_{r=1}^{R_i} (\bar{y}_{irs} - y_{ir})^2}, RMSE = \frac{1}{NS} \sum_{i=1}^N \sum_{s=1}^S e_{is} \quad (14)$$

where R_i is the number of RPTs of cell i , $N=64$ and $S=3$. The mean absolute error (MAE) is aggregated in the same way, and errors are expressed in SOH percentage points (pp). Uncertainty is estimated by 5,000 fold-stratified cell bootstrap resamples that preserve the pairing of cells between models. Paired differences are defined as $\Delta RMSE = RMSE_{\text{candidate}} - RMSE_{\text{reference}}$, so negative values favour the candidate, and p values come from 5,000 two-sided paired sign-flip randomisations. Holm correction is applied within nine prespecified test families at $\alpha = 0.05$ (Supplementary Note S4); stratified, bias, marginal and post hoc analyses are descriptive. Intervals are conditional on the fixed split and trained models.

4. Results and discussion

Unless stated otherwise, RMSE and MAE follow Eq. (14), intervals are 95% fold-stratified cell bootstrap intervals, and “improved cells” counts cells with lower error in a paired comparison.

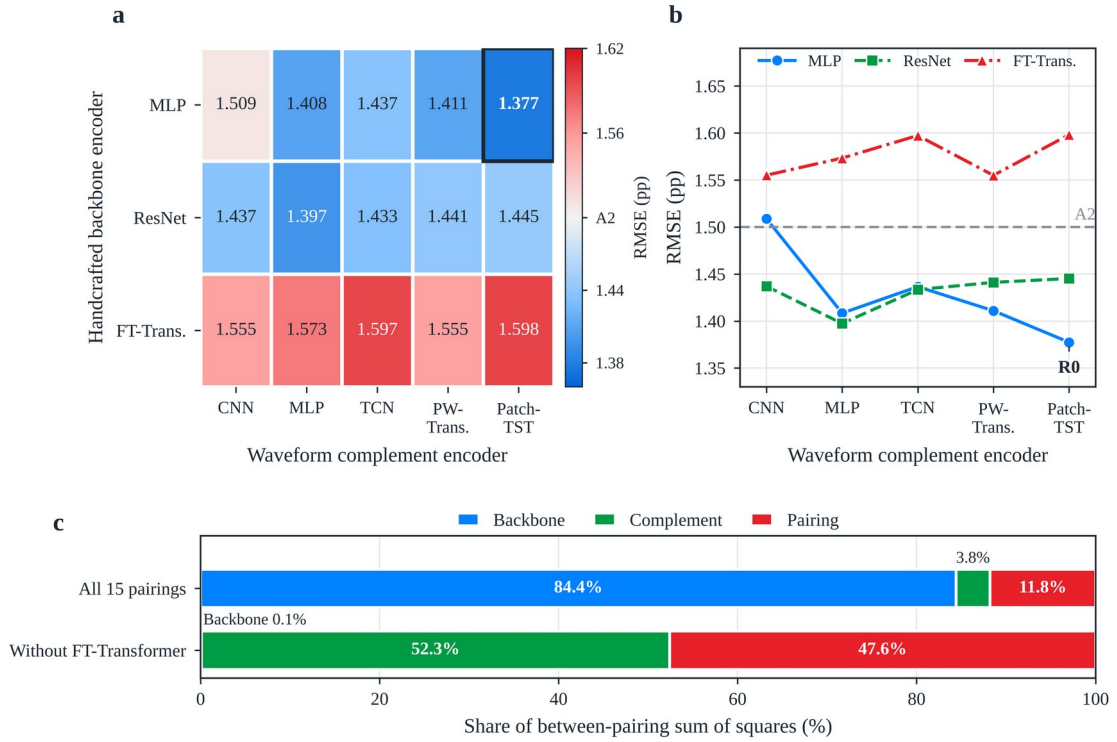
4.1. Encoder pairings and their dependence

Among the 15 pairings, R0 (MLP backbone + PatchTST complement) attains the lowest RMSE, 1.377 pp (1.247–1.513), with an MAE of 1.141 pp (Table 4, Fig. 3a). Relative to the MLP + CNN pairing it reduces RMSE by 0.131 pp (−0.160 to −0.101; Holm $p = 0.003$), or 8.7%, improving 51 of 64 cells. The runner-up, ResNet + MLP, reaches 1.397 pp, 0.020 pp above R0; because not all pairs were tested against each other, the lowest observed mean is not claimed to be significantly better than every alternative. R0 also ranks first on the validation cells (1.310 pp), and the top three pairings and their order coincide between validation and test; across the five test folds R0 ranks 8th, 1st, 1st, 1st and 3rd.

At the level of single factors, the FT-Transformer backbone raises the error by 0.147 pp on average relative to the MLP (0.107–0.191), whereas ResNet and MLP are indistinguishable; the average effects of the waveform encoders are small (−0.041 to −0.011 pp relative to the CNN). The ranking of complement encoders, however, depends on the backbone (Fig. 3b). PatchTST is best with the MLP backbone but last with ResNet and FT-Transformer, which prefer the MLP and point-wise Transformer complements, respectively. Taking MLP + CNN as the double reference, all eight difference-in-differences are positive (ResNet 0.061–0.140 pp, FT-Transformer 0.098–0.174 pp; uncorrected $p \leq 0.0022$), so the advantage of a complement over the CNN shrinks or reverses when the backbone changes.

A sum-of-squares decomposition of the 3×5 RMSE table attributes 84.4% of the between-pairing variation to the backbone, 3.8% to the complement and 11.8% to

531 their pairing (Fig. 3c). Without the FT-Transformer row the shares become 0.1%,
532 52.3% and 47.6%, so the apparent backbone dominance reflects the higher error of
533 FT-Transformer, while the pairing term is substantial in both cases. Encoders should
534 therefore be selected as pairings on controlled evidence, not by architectural
535 complexity; this agrees with the BMS preference for lightweight, interpretable models
536 [18,27] and supports the choice of an MLP backbone with a PatchTST complement.



537 **Fig. 3.** Main effects and pairing dependence in the encoder matrix. (a) Cell-macro RMSE of
538 the 15 pairings; the diverging scale is centred on the backbone-only model A2 (1.500 pp),
539 blue denoting lower and red higher error; the frame marks R0. (b) Interaction plot; non-
540 parallel lines indicate pairing dependence; the dashed line marks A2. (c) Share of the
541 between-pairing sum of squares due to backbone, complement and pairing for all 15 pairings
542 (95% intervals: 73.1–90.2%, 1.9–8.1%, 7.3–20.2%) and without FT-Transformer (0.0–4.6%,
543 40.4–65.0%, 34.1–58.4%), from 5,000 fold-stratified cell bootstraps. Values are seed-
544 averaged cell metrics from 225 trainings.

546 **Table 4.** Errors of all 15 encoder pairings (SOH pp).

Backbone	Compleme nt	Ran k	RMSE	95% CI	MAE	Val. RMSE (rank)	Δ RMSE _a	Holm p
MLP	PatchTST	1	1.377	1.247–1.513	1.141	1.310 (1)	–0.131	0.003
MLP	MLP	3	1.408	1.271–1.553	1.154	1.321 (3)	–0.100	0.003

Backbone	Complement	Rank	RMSE	95% CI	MAE	Val. RMSE (rank)	Δ RMSE ^a	Holm p
MLP	PW-Trans.	4	1.411	1.276–1.546	1.164	1.341 (5)	−0.098	0.003
MLP	TCN	6	1.437	1.294–1.585	1.175	1.356 (8)	−0.072	0.003
MLP	CNN	10	1.509	1.368–1.655	1.249	1.408 (10)	—	—
ResNet	MLP	2	1.397	1.258–1.548	1.152	1.312 (2)	−0.111	0.003
ResNet	TCN	5	1.433	1.296–1.581	1.186	1.332 (4)	−0.075	0.003
ResNet	CNN	7	1.437	1.304–1.578	1.200	1.367 (9)	−0.072	0.003
ResNet	PW-Trans.	8	1.441	1.303–1.583	1.199	1.355 (7)	−0.068	0.003
ResNet	PatchTST	9	1.445	1.309–1.592	1.200	1.352 (6)	−0.064	0.003
FT-Transformer	PW-Trans.	11	1.555	1.406–1.709	1.303	1.495 (12)	+0.046	0.103
FT-Transformer	CNN	12	1.555	1.404–1.710	1.305	1.482 (11)	+0.046	0.093
FT-Transformer	MLP	13	1.573	1.420–1.731	1.315	1.495 (13)	+0.065	0.038
FT-Transformer	TCN	14	1.597	1.448–1.758	1.333	1.518 (14)	+0.089	0.003
FT-Transformer	PatchTST	15	1.598	1.446–1.757	1.340	1.525 (15)	+0.089	0.003

^a Pairing minus MLP + CNN; negative values denote lower error. Holm p values refer to the 14 comparisons against MLP + CNN, not to comparisons between pairings. Validation RMSE is the mean best-epoch error on validation cells over 15 runs.

4.2. Waveform increment over the handcrafted backbone

Adding the PatchTST branch to the MLP backbone lowers the error on the same test cells (Table 5, Fig. 4a). R0 improves on A2 by 0.123 pp (−0.141 to −0.103; Holm $p < 0.001$), with 58 of 64 cells improved and a paired effect size $d_z = -1.55$. Relative to A1, which lacks the handcrafted backbone, R0 improves by 0.117 pp (−0.187 to −0.033; Holm $p = 0.002$) in 50 of 64 cells, with a larger spread between cells ($d_z = -0.36$, Fig. 4b). A1 and A2 do not differ (−0.006 pp, −0.091 to 0.065; $p = 0.91$); because A1 still contains the DCIR and τ , this is not a pure waveform-versus-feature comparison. The two representations are thus complementary on average.

Table 5. Ablation of the backbone, waveform complement, cycle count and monotonicity penalty (SOH pp).

Config.	Inputs and constraint	RMS E	MAE	Contrast	Δ RMS E	95% CI	Holm p	Improved cells
A2	Backbone + τ , $\lambda = 0.1$	1.500	1.244	—	—	—	—	—
R0	Backbone + PatchTST + τ , $\lambda = 0.1$	1.377	1.141	R0 − A2	−0.123	−0.141 to −0.103	<0.001	58/64

Config.	Inputs and constraint	RMS E	MAE	Contrast	Δ RMS E	95% CI	Holm p	Improved cells
A1	PatchTST + DCIR + τ , $\lambda = 0.1$	1.494	1.259	R0 – A1	-0.117	-0.187 to -0.033	0.002	50/64
A3	As R0, $\lambda = 0$	1.406	1.164	R0 – A3	-0.028	-0.038 to -0.018	<0.001	46/64
A4	As A3, without τ	1.983	1.718	A3 – A4	-0.578	-0.731 to -0.421	<0.001	52/64

Δ RMSE is the first minus the second configuration of each contrast; improved cells are those where the first has the lower error. The four contrasts are prespecified confirmatory tests with Holm correction among them.

Not every waveform encoder delivers this increment (Table 6, Fig. 4c). With the MLP backbone fixed, PatchTST, MLP, point-wise Transformer and TCN complements lower RMSE by 0.123, 0.092, 0.089 and 0.063 pp (Holm $p < 0.001$ for all), whereas the CNN complement changes it by +0.009 pp (-0.012 to 0.028; Holm $p = 0.491$). PatchTST gives the largest observed reduction, consistent with its rank in Section 4.1.

Table 6. Increment of each waveform complement over the MLP backbone A2 (SOH pp).

Complement	Dual-branch RMSE	Δ RMSE	95% CI	Relative change	Holm p	Improved cells	d_z
PatchTST	1.377	-0.123	-0.141 to -0.104	-8.2%	<0.001	58/64	-1.55
MLP	1.408	-0.092	-0.109 to -0.074	-6.1%	<0.001	55/64	-1.14
PW-Trans.	1.411	-0.089	-0.109 to -0.070	-5.9%	<0.001	54/64	-0.97
TCN	1.437	-0.063	-0.089 to -0.037	-4.2%	<0.001	45/64	-0.56
CNN	1.509	+0.009	-0.012 to 0.028	+0.6%	0.491	28/64	+0.09

Δ RMSE is the dual-branch model minus A2 (1.500 pp); d_z is the mean per-cell difference divided by its standard deviation. Holm correction covers these five comparisons.

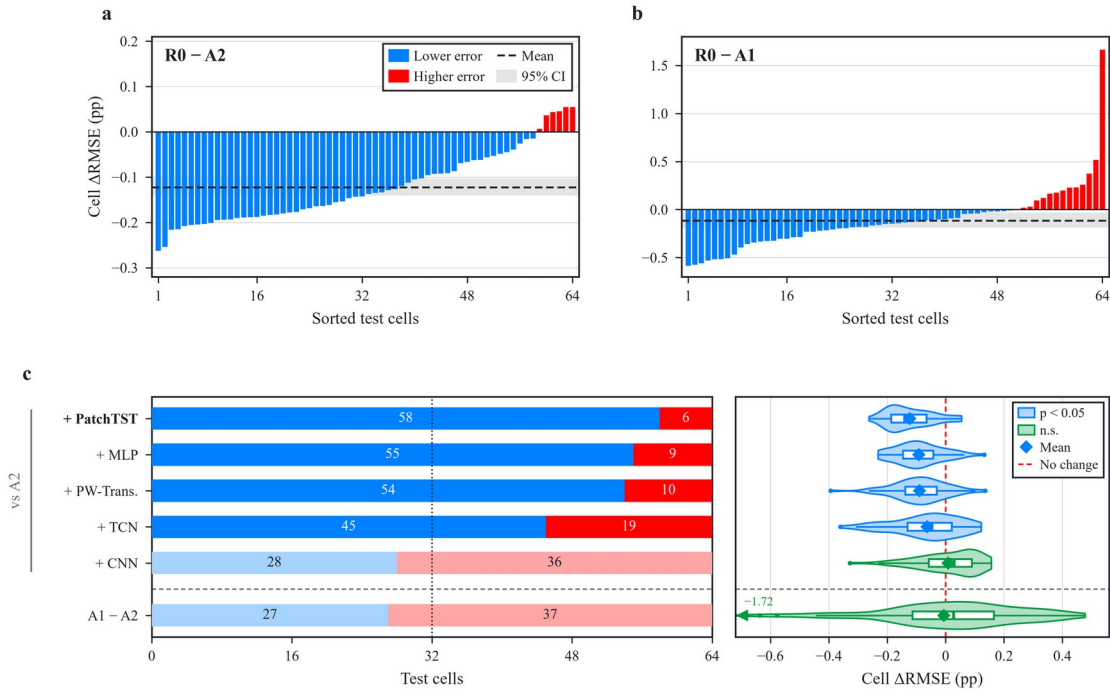
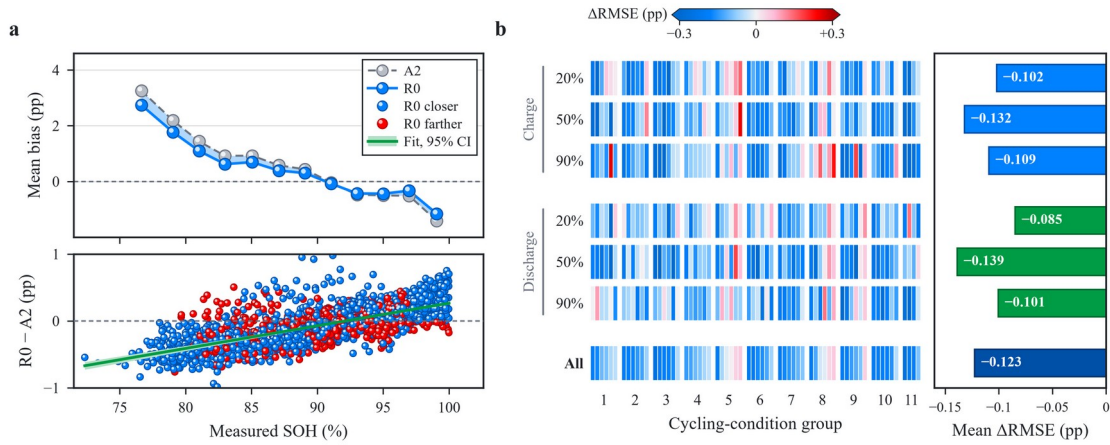


Fig. 4. Effects of the handcrafted backbone and the waveform complement. (a) Per-cell RMSE difference R0 - A2 (seed-averaged) for the 64 test cells sorted in ascending order; blue bars denote lower and red bars higher error for R0; the dashed line and grey band show the mean difference and its 95% interval. (b) As (a) for R0 - A1; the rightmost bar is cell 3. (c) Left: numbers of cells with lower (blue) or higher (red) error after adding each complement to the MLP backbone, and for A1 versus A2 (below the dashed line); pale colours mark non-significant differences. Right: violin and box plots of the 64 paired differences (box, interquartile range; line, median; whiskers, $1.5 \times$ IQR; diamond, mean); one value outside the axis (A1 - A2, -1.72 pp, cell 3) is marked at the left edge.

4.3. Distribution of the increment and bias correction

The increment varies in size across SOH ranges but has the same sign in all six pulse conditions (Supplementary Table S4, Fig. 5b). Below 80% SOH, R0 lowers the error relative to A2 by 0.413 pp (17.2%, 62/64 cells), and above 95% by 0.148 pp (15.5%, 61/64); the 90–95% range shows the smallest gain (0.027 pp, 36/64). Across the six conditions the reduction ranges from 0.085 to 0.139 pp; 313 of the 384 cell-condition units (81.5%) improve, 29 cells improve in all six conditions, and all 11 cycling groups show negative mean differences (−0.029 to −0.186 pp). Each single pulse condition therefore benefits from the complement, which matters for deployments that can afford only one pulse.

593 Binned bias analysis shows that the increment partly corrects a shrinkage bias of
 594 the backbone (Fig. 5a). Over the 2,823 cell-RPT points, A2 overestimates SOH by
 595 3.25 pp below 78% and underestimates it by 1.41 pp at 98–100%; R0 reduces these to
 596 2.74 and 1.16 pp. The change in prediction from A2 to R0 increases with true SOH
 597 (slope 0.034 pp per pp, 0.031–0.036) and crosses zero near 92.1%: predictions are
 598 lowered at low SOH and raised at high SOH, opposite to the bias of A2. As a result,
 599 68.0% of the points move closer to the measurement, the slope of predicted on true
 600 SOH rises from 0.848 to 0.882, and the total squared error falls by 14.4%.



601 **Fig. 5.** Bias correction by the waveform complement and its distribution across conditions
 602 and cells. (a) Top: mean bias (prediction minus truth) in 2-pp bins of measured SOH (bins
 603 below 78% merged) for A2 (grey) and R0 (blue); the shading marks the corrected bias.
 604 Bottom: R0 - A2 prediction change at the 2,823 cell-RPT points (blue, closer to the
 605 measurement; red, farther) with the least-squares fit and its 95% bootstrap band. Predictions
 606 are averaged over seeds and conditions. (b) Left: per-cell R0 - A2 RMSE difference for the
 607 six conditions and all records (cells ordered by cycling group; colours saturate beyond ± 0.3
 608 pp). Right: cell-macro mean difference per row; all 95% intervals exclude zero.

610 4.4. Learner dependence and comparison with reference methods

611 The comparison with reference methods delimits where the dual-branch design
 612 helps (Table 7; all results in Supplementary Table S5). The reproduced raw-pulse
 613 model of Nowacki et al. [28] reaches 1.628 pp, and R0 is 0.251 pp lower (Holm p =
 614 0.019); since that model uses only the raw pulse without τ , the gap cannot be

attributed to the dual-branch structure alone. On the A2 input, GBDT and RBF-SVR reach 1.022 and 1.026 pp. A2 and GBDT share the same input yet differ by about 0.48 pp, so the backbone learner matters on these data. Models using τ alone err by about 3 pp and GBDT without τ reaches 1.663 pp, so the cycle record and the pulse features both contribute.

Table 7. Inputs and cell-macro errors of representative methods (SOH pp).

Model	Input	RMSE	95% CI	Δ RMSE vs. A2	Δ RMSE vs. R0	Holm p^a	Cells better than R0
R0 (this work)	Features + τ + ΔV	1.377	1.247–1.513	−0.123	—	—	—
A2 (backbone only)	Features + τ	1.500	1.365–1.637	—	+0.123	<0.001	6/64
Nowacki et al. [28], reproduced	Raw ΔV	1.628	1.454–1.820	+0.128	+0.251	0.019	17/64
GBDT	Features + τ	1.022	0.897–1.160	−0.478	−0.355	0.006	58/64
RBF-SVR	Features + τ	1.026	0.915–1.146	−0.474	−0.351	0.006	57/64

^a Versus R0. R0 versus A2 belongs to the prespecified family of Table 5; the other comparisons belong to the 30-comparison extended family (Supplementary Note S4).

The waveform increment does not carry over to the GBDT backbone (Fig. 6). Whereas the neural backbone gains 0.123 pp in 58 of 64 cells, $(G+R0) - (G+A2)$ is only −0.008 pp with 36 of 64 cells improved (Holm $p = 0.35$); none of the three residual learners improves GBDT significantly, and appending the raw waveform to GBDT gives no gain either (Supplementary Note S5). This pattern is consistent with the waveform branch recovering information that the neural backbone leaves unexploited in the handcrafted features, which a tree ensemble already extracts. The practical rule is therefore that the waveform complement should be added when a neural handcrafted backbone is used, for example where end-to-end differentiability is needed for derivative constraints, while a well-tuned tree ensemble can rely on the handcrafted features.

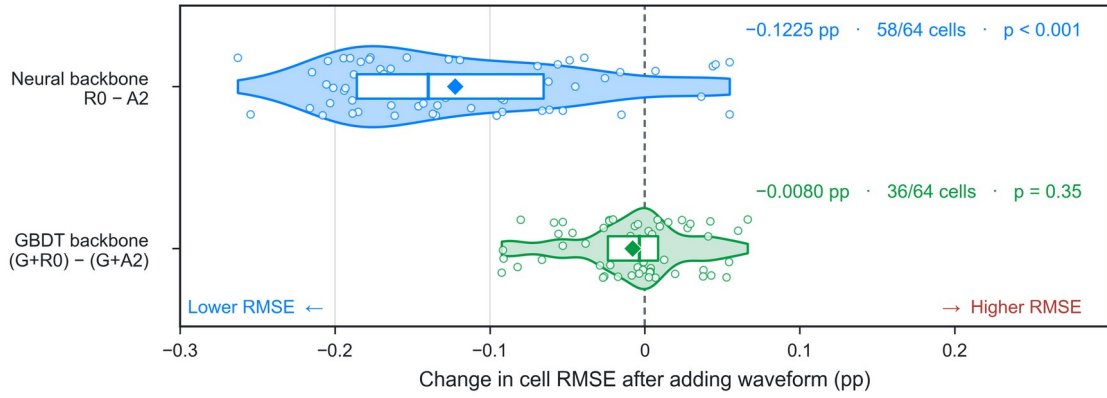


Fig. 6. Dependence of the waveform increment on the backbone learner. Paired change in cell RMSE after adding the waveform (negative = lower error) for the 64 test cells. Top: neural backbone, R0 – A2. Bottom: GBDT backbone with the waveform added through residual networks, (G+R0) – (G+A2). Circles, cells; violins, distributions; boxes, interquartile ranges with medians; diamonds, means. Annotations give the mean difference, improved cells and Holm-adjusted p .

4.5. Contribution of design factors

4.5.1. Cumulative cycle information and monotonicity penalty

Among the ablated factors, removing the cycle count increases the error most (Table 5, Fig. 7a–c). Without the monotonicity penalty in either model, A3 (with τ) is 0.578 pp below A4 (without τ ; reduction 0.421–0.731; Holm $p < 0.001$) in 52 of 64 cells, i.e. removing τ raises the error by about 41%. Together with the large error of τ -only models (Section 4.4), the cycle record is an important but not self-sufficient input.

The soft monotonicity penalty yields a small accuracy gain (Fig. 7d–f): R0 is 0.028 pp below A3 (reduction 0.018–0.038; Holm $p < 0.001$) in 46 of 64 cells. The share of positive partial derivatives with respect to τ (5.39% for A3, 5.61% for R0) and of rising predictions between adjacent RPTs of the same cell and condition (26.88% and 27.03%) remain essentially unchanged, so the penalty acts as a regulariser rather than as a guarantee of monotone trajectories.

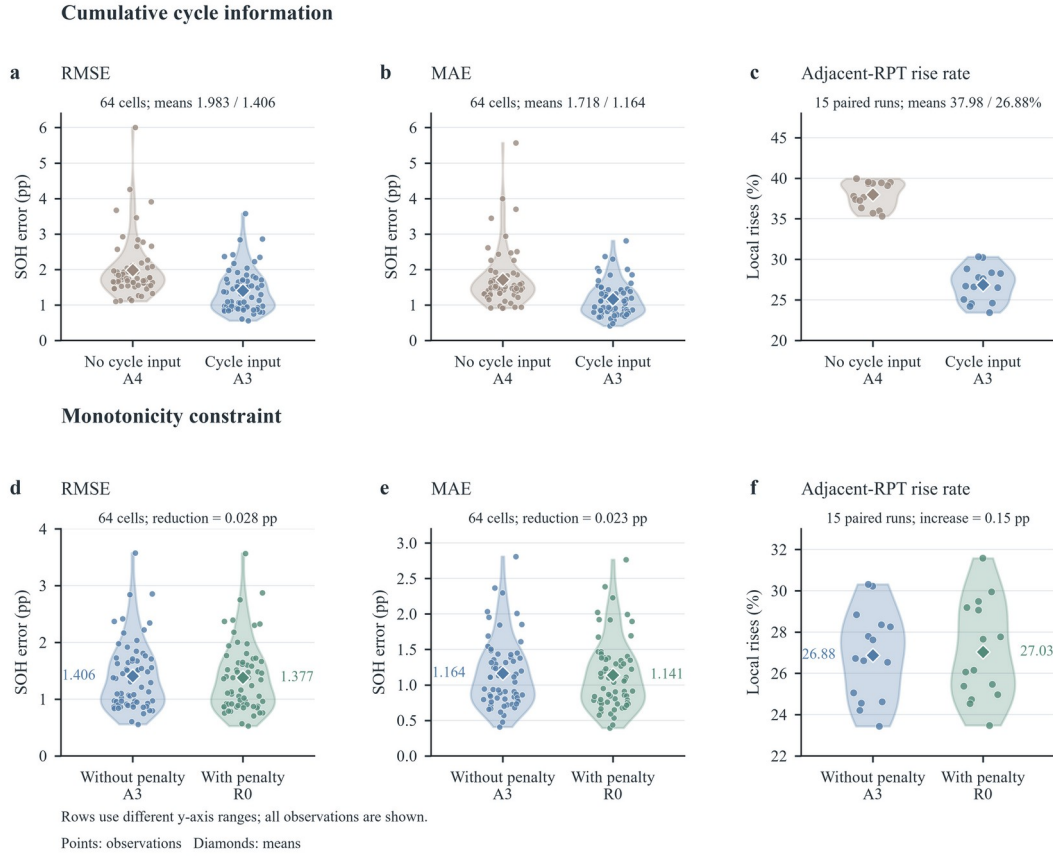


Fig. 7. Effects of cumulative cycle information and the monotonicity penalty. Top row: A4 (no cycle input) versus A3 (cycle input), both without the penalty. Bottom row: A3 (no penalty) versus R0 (penalty). (a, d) RMSE; (b, e) MAE; (c, f) rate of rising predictions between adjacent RPTs. Error distributions contain the 64 test cells (seed-averaged); rise rates are computed within cell, nominal SOC and direction for the 15 runs. Violins are kernel densities limited to the observed range; points are observations and diamonds means. Rows use different y-axis ranges.

4.5.2. Design of the waveform branch

On the 2×2 patch grid, R0's (9, 4) setting gives the lowest error (Table 8; Supplementary Fig. S1). P1 (5, 2), P2 (5, 4) and P3 (9, 2) raise RMSE by 0.072, 0.094 and 0.055 pp (Holm $p < 0.001$), and length and stride interact: stride 4 is better with length 9 but stride 2 is better with length 5 (difference-in-differences -0.077 pp, -0.100 to -0.054). Of the three adaptations, the mean-pooling head is clearly supported: reverting to a flatten head (D1) raises the error by 0.062 pp (0.036–0.089; Holm $p < 0.001$) while more than doubling the parameter count (19,041 to 42,593).

671 Adding RevIN (D2, -0.020 pp, -0.049 to 0.005) or using the raw voltage (D3, $+0.007$
672 pp, -0.005 to 0.017) makes no detectable difference (Holm $p = 0.345$ for both). With
673 A2 as the common reference, all six PatchTST variants reduce the error by 0.028 –
674 0.143 pp (40–57 of 64 cells improved), so the increment is not tied to a single patch
675 setting.

676 **Table 8.** PatchTST patch parameters and adaptation ablations (SOH pp).

Config	Difference from R0	RMS E	95% CI	Params	Δ RMS E	95% CI of Δ RMSE	Holm p	R0 better
R0	Patch (9, 4), mean-pooling head, no RevIN, ΔV input	1.377	1.247–1.513	19,041	—	—	—	—
P1	Patch (5, 2)	1.449	1.315–1.585	19,713	+0.072	0.050–0.094	<0.001	51/64
P2	Patch (5, 4)	1.472	1.341–1.611	18,945	+0.094	0.080–0.109	<0.001	56/64
P3	Patch (9, 2)	1.432	1.297–1.570	19,777	+0.055	0.038–0.071	<0.001	47/64
D1	Flatten head	1.440	1.304–1.585	42,593	+0.062	0.036–0.089	<0.001	46/64
D2	With RevIN	1.357	1.229–1.491	19,043	-0.020	-0.049 to 0.005	0.345	28/64
D3	Raw voltage input	1.384	1.253–1.518	19,041	$+0.007$	-0.005 to 0.017	0.345	41/64

677 Δ RMSE is the configuration minus R0; positive values mean the corresponding R0 design is beneficial.
678 Patch (P1–P3) and adaptation (D1–D3) ablations form two families with Holm correction over three
679 comparisons each.

680 4.5.3. Number of handcrafted features

681 Within the present candidate set, $K=50$ is supported (Supplementary Table S6,
682 Fig. S2). $K = 10, 20$ and 30 raise RMSE by $0.139, 0.122$ and 0.091 pp (Holm $p =$
683 0.001), $K = 75$ and 100 are slightly worse ($+0.037$ and $+0.041$ pp; Holm $p = 0.002$),
684 and $K=144$ is indistinguishable from $K=50$ (-0.002 pp, -0.019 to 0.015 ; Holm $p =$
685 0.838) while requiring 31.6% more parameters. $K=50$ is therefore a balanced choice
686 between accuracy and input size.

687 4.6. Generalisation, robustness and deployment

688 Except for the main NMC/graphite results, which come from the original
689 implementation, the experiments in this section use the CPU re-implementation of

690 Section 3.5.5, which reproduces the cleaning exactly and the reported errors closely,
691 including the sign and significance of the waveform increment ($R0 - A2 = -0.046$ pp,
692 -0.070 to -0.023 ; $p < 0.001$; 42 of 64 cells) and the identity of the best- and worst-
693 estimated cells (cells 12 and 3). All comparisons below are made within this
694 implementation (Table 9, Fig. 8).

695 *Unseen cycling conditions.* When each cycling group is held out entirely, the
696 error of the neural models rises only moderately ($R0$ from 1.427 to 1.560 pp, +9.3%;
697 $A2$ +8.2%), whereas that of GBDT rises sharply (from 1.028 to 1.401 pp, +36.2%), so
698 the advantage of GBDT over $R0$ shrinks from 0.40 to 0.16 pp (Fig. 8a). For group 3,
699 the condition hardest to extrapolate, $R0$ is 0.59 pp more accurate than GBDT. The
700 waveform increment persists for unseen conditions ($R0 - A2 = -0.035$ pp, -0.053 to
701 -0.015 ; $p = 0.008$; 39 of 64 cells), with lower group-mean errors in seven of the 11
702 held-out groups.

703 *Independent chemistry.* On the NMC/graphite dataset (original implementation,
704 five folds $\times$ three seeds), $R0$ reaches a cell-macro RMSE of 1.968 pp (95% CI 1.728–
705 2.236) against 2.132 pp for $A2$: $R0 - A2 = -0.163$ pp (Holm-adjusted $p = 0.0004$),
706 with 39 of 44 cells improved and a 7.6% error reduction, close to the 8.2% on LFP;
707 the difference is negative in every outer fold and every initialisation seed, and $A1$,
708 without the handcrafted branch, reaches 2.468 pp. The waveform increment is
709 therefore reproduced on a second electrode chemistry. As on LFP, GBDT is more
710 accurate on these data (0.965 pp, better in 43 of 44 cells), with a larger gap that arises
711 mainly from the shrinkage of the neural models at both ends of the SOH range: below
712 60% SOH, $R0$ overestimates by 3.3 pp on average (GBDT 1.0 pp), and at or above
713 95% SOH it underestimates by 1.75 pp; the waveform branch also reduces the low-
714 SOH overestimation ($A2$ 4.1 pp). A GBDT using only τ errs by 8.8 pp, whereas one

using only the pulse features without τ reaches 1.28 pp, so the SOH information on NMC comes mainly from the pulse itself.

Information added by the waveform branch. The open-circuit voltage of NMC is sloped, so capacity fade shifts the actual SOC at which pulses are applied and the pre-pulse voltage itself tracks SOH: its absolute Spearman correlation with SOH averages 0.93 over the six conditions, against 0.36 for LFP. The waveform branch receives the first-sample-referenced voltage ΔV , which excludes this absolute level, and supplies the dynamic response during the pulse. When the handcrafted features include absolute-level quantities such as the pre-pulse voltage or voltage quantiles, the two sources partly overlap: in the re-implementation, the NMC waveform increment vanishes when such features enter the backbone (+0.004 pp) and returns to -0.158 pp (-0.205 to -0.112; $p < 0.001$; 36 of 44 cells) when they are removed from the candidates, matching the original implementation (Supplementary Note S6). The waveform branch thus contributes dynamic information beyond the absolute voltage level rather than repeating what the handcrafted features already encode.

Cycle-count errors and sensor noise. A systematic $\pm 10\%$ error in the cycle count raises R0's error by only 1.5–3.5% (1.449 and 1.477 pp), compared with 8.6–14.4% for GBDT (1.117 and 1.176 pp), and leaves the waveform increment intact (-0.048 and -0.041 pp; $p \leq 0.002$) (Fig. 8b). With matched sensor noise in training and test data, accuracy degrades gracefully: at $\sigma_{\text{noise}} = 1$ and 2 mV, R0 reaches 1.620 and 1.715 pp (+13.5% and +20.2%), against +33.5% and +38.5% for GBDT (1.372 and 1.424 pp), and the waveform increment is maintained (-0.058 pp in 50 of 64 cells and -0.045 pp in 46 of 64 cells; $p \leq 0.002$).

Number of pulses. With a single pulse per RPT, R0 reaches 1.549 pp on average over the six conditions (1.490–1.582 pp); two and three pulses give 1.477 and 1.453

740 pp, against 1.427 pp for all six (Fig. 8c), and R0 stays below A2 for every pulse count.
741 The three charge pulses alone reach 1.432 pp, within 0.005 pp of the full protocol, so
742 the diagnostic time can be halved, and the pulses can be applied during charging as
743 proposed for vehicles [16].

744 *Attribution.* Integrated gradients spread R0’s waveform attribution almost evenly
745 over the response: the mean absolute attribution per sample is 0.0132, 0.0138 and
746 0.0135 pp in the C/5, 1C and rest segments, with the same profile for all six
747 conditions (Fig. 8d). The complement therefore draws on the shape of the whole
748 transient rather than on one segment that a single handcrafted descriptor could isolate,
749 in line with its intended role of capturing inter-segment information.

750 *Computational cost.* R0 has 19,041 parameters (74 kB in 32-bit floating point)
751 and needs 0.42 MFLOPs per pulse, most of them in the waveform branch (A2: 13.7
752 kFLOPs). On a single CPU thread, one inference takes 0.52 ms in PyTorch and the
753 reference Python implementation of all 143 waveform features takes 6.5 ms, both
754 negligible against the 100-s pulse, whereas the validation-selected GBDT comprises
755 1,000 trees with 61,000 nodes. This footprint is of the order targeted by embedded
756 SOH estimators [19].

757 **Table 9.** Generalisation and robustness in the re-implementation (cell-macro RMSE, SOH
758 pp).

Scenario	R0	A2	GBDT	R0 – A2	95% CI	<i>p</i>	Improve d cells
Five-fold cell-disjoint (reference)	1.427	1.474	1.028	−0.046	−0.070 to −0.023	<0.001	42/64
Leave-one-group-out	1.560	1.594	1.401	−0.035	−0.053 to −0.015	0.008	39/64
Cycle count +10% (test)	1.449	1.497	1.117	−0.048	−0.072 to −0.024	<0.001	43/64
Cycle count −10% (test)	1.477	1.518	1.176	−0.041	−0.063 to −0.018	0.002	39/64
Matched noise, $\sigma_{\text{noise}} = 1$ mV	1.620	1.678	1.372	−0.058	−0.077 to −0.040	<0.001	50/64

Scenario	R0	A2	GBDT	R0 - A2	95% CI	p	Improve d cells
Matched noise, $\sigma_{\text{noise}} = 2$ mV	1.715	1.760	1.424	-0.045	-0.069 to -0.022	0.002	46/64
One pulse per RPT ^a	1.549	1.574	1.203	—	—	—	—
Three pulses per RPT ^a	1.453	1.494	1.066	—	—	—	—
Three charge pulses	1.432	—	—	—	—	—	—

Paired comparisons use 5,000 fold- (or group-) stratified cell bootstraps and sign-flip tests; descriptive, without multiplicity correction. ^a Mean over all condition combinations (6 and 20, respectively).

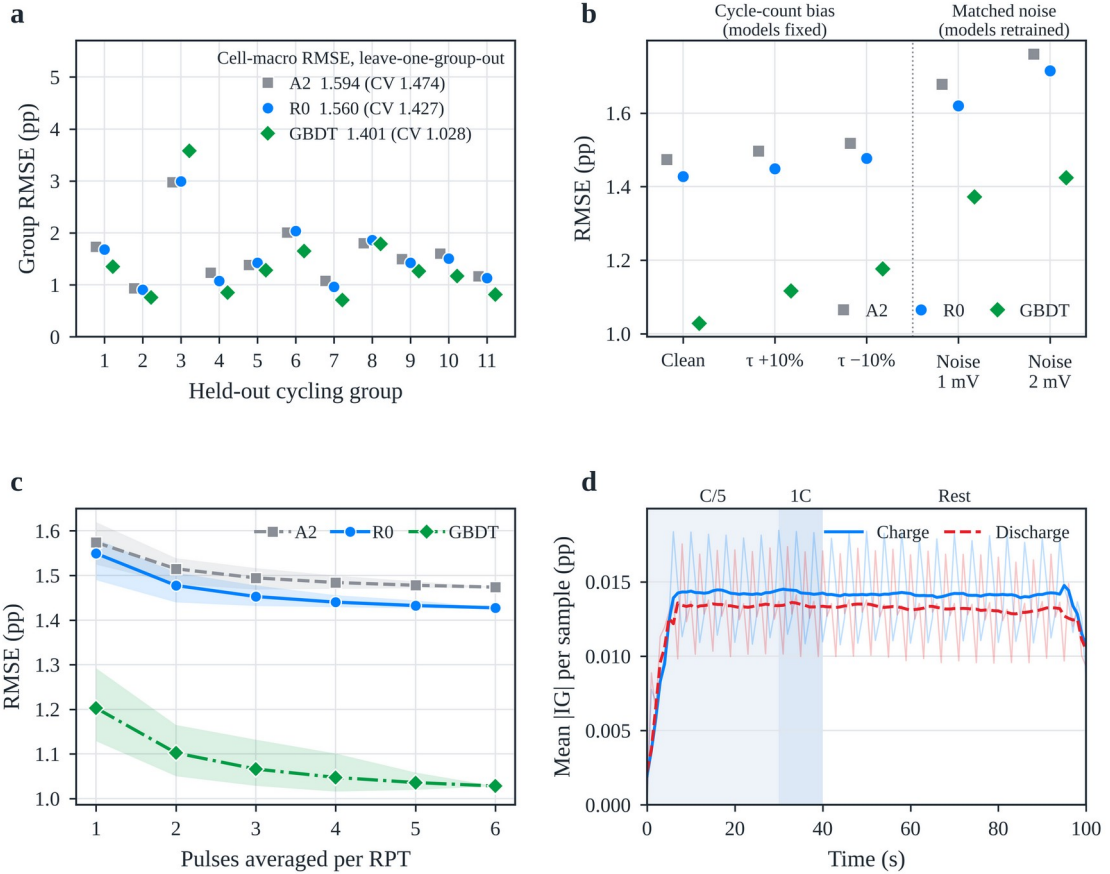


Fig. 8. Generalisation, robustness and attribution (re-implementation). (a) Group-mean RMSE when each cycling group is held out (leave-one-group-out); the legend gives the cell-macro RMSE under leave-one-group-out and, in parentheses, under five-fold cell-disjoint validation. (b) RMSE with a $\pm 10\%$ bias of the test cycle count (models fixed) and with matched Gaussian voltage noise in training and test data (models retrained). (c) RMSE versus the number of pulses averaged per RPT; lines show the mean over all condition combinations and bands their range. (d) Mean absolute integrated-gradient attribution of R0 per waveform sample for charge (solid) and discharge (dashed) pulses, averaged over test records, folds and seeds; shading marks the pulse segments.

4.7. Error characteristics and scope

The overall mean bias is close to zero but conceals SOH-dependent errors (Supplementary Fig. S3). Averaging the three initialisations of each record, R0 reaches a record-level RMSE of 1.461 pp, an MAE of 1.089 pp and a mean bias of -0.093 pp over the 16,938 records. The 630 records below 80% SOH are overestimated by 2.003 pp on average and the 5,682 records above 95% are underestimated by 0.574 pp (GBDT: +1.068 and -0.130 pp). Individual cells differ markedly (Supplementary Fig. S4): cells 12, 6, 44 and 3, ranked 1st, 22nd, 43rd and 64th by R0 error, reach 0.526, 0.961, 1.519 and 3.568 pp (A2: 0.704, 1.066, 1.723 and 3.624 pp). Across the 11 cycling groups the cell-macro RMSE spans 0.878–2.333 pp (Supplementary Fig. S5), with positive biases in groups 3 and 6 and negative biases in group 8; the correlation between the per-cell fade rate and the mean bias is 0.59.

The analysis treats the 64 cells as the unit of inference on a fixed split; the low-SOH overestimation is reduced but not removed, and training data should come from the target sensor because difference- and spectrum-based features are sensitive to noise absent from training (Supplementary Note S6). Validation on field data with varying temperature and on target BMS hardware remains future work.

5. Conclusions

Using 16,938 short-pulse records from 64 public LFP/graphite cells, 15 encoder pairings were compared under one five-fold cell-disjoint protocol and training budget. The main conclusions are as follows.

- (1) Encoders should be evaluated as pairings. The MLP backbone with a PatchTST complement (R0) gives the lowest observed RMSE (1.377 pp), 8.7% below MLP + CNN, while PatchTST ranks last with the other two backbones; the pairing term explains 11.8% of between-pairing variation (47.6% without FT-Transformer).

797 (2) The waveform complement improves a neural handcrafted backbone. R0 is
798 0.123 pp (8.2%) below the backbone-only model in 58 of 64 cells, with the
799 largest gain below 80% SOH (17.2%), consistent improvements in all six
800 pulse conditions and a partial correction of the backbone’s shrinkage bias.
801 With a GBDT backbone, which already reaches 1.022 pp, the waveform adds
802 no measurable gain.

803 (3) The waveform increment is reproduced on an independent NMC/graphite
804 dataset ($R0 - A2 = -0.163$ pp, 7.6%, 39 of 44 cells) and originates from the
805 dynamic response beyond the absolute voltage level. In a re-implementation, it
806 also persists for held-out cycling conditions (-0.035 pp) and under matched
807 sensor noise of 1–2 mV (-0.045 to -0.058 pp), and the neural model degrades
808 far less than GBDT when cycling conditions are unseen (+9% versus +36%)
809 or the cycle count is biased by $\pm 10\%$ (at most +3.5% versus +14.4%).

810 (4) Among the ablated factors, the cumulative cycle count is the most influential
811 input (+41% error without it), and the monotonicity penalty acts as a mild
812 regulariser (-0.028 pp). The (9, 4) patch setting and the mean-pooling head
813 (-0.062 pp and 55% fewer parameters than a flatten head) are the most
814 effective design choices, and $K=50$ balances accuracy and input size.

815 (5) The model is light enough for embedded use, with 19,041 parameters (74 kB),
816 0.42 MFLOPs and sub-millisecond single-core inference per pulse; its
817 attribution is spread over the whole transient, and the three charge pulses alone
818 match the six-pulse protocol within 0.005 pp.

819 Nomenclature

Abbreviation	Definition	Symbol	Definition
BMS	Battery management system	E_h, E_v	Backbone and complement encoders
CNN	Convolutional neural network	g	Regression head
DCIR	Direct-current internal resistance	K	Number of selected handcrafted features
DOD	Depth of discharge	L, P, S	Sequence length, patch length, patch stride
GBDT	Gradient-boosting decision tree	M	Number of patches
GPR	Gaussian process regression	Q_{ir}	Discharge capacity of cell i at RPT r
LFP	Lithium iron phosphate	y_{ir}	Measured SOH (%)
MAE	Mean absolute error	$\hat{y}, \hat{y}^*$	Predicted SOH and its standardised value
MLP	Multilayer perceptron	$\Delta v_t, \tilde{v}_t$	Referenced and standardised pulse voltage
NMC	Nickel–manganese–cobalt oxide	h	Backbone input vector
PW-Trans.	Point-wise Transformer	$n_{i,r}, \tau$	Cumulative cycle count and its standardised value
RBF-SVR	Radial-basis-function support vector regression	λ	Weight of the monotonicity penalty
RevIN	Reversible instance normalisation	ρ_j	Spearman correlation of feature j with SOH
RMSE	Root-mean-square error	σ_{noise}	Standard deviation of added voltage noise
RPT	Reference performance test	R0	Reference configuration (MLP + PatchTST)
SOC, SOH	State of charge, state of health	A1–A4, D1–D3, P1–P3	Ablation configurations (Section 3.5.1)
TCN	Temporal convolutional network	pp	Percentage points of SOH

820 Declaration of Competing Interest

821 The authors declare that they have no known competing financial interests or
822 personal relationships that could have appeared to influence the work reported in this
823 paper.

824 Data availability

825 The battery data are publicly available from the UConn-ISU-ILCC LFP and
826 UConn-ILCC NMC ageing datasets [28,31]. Cleaning rules, input definitions, splits
827 and metrics are specified in Sections 2 and 3 and Supplementary Notes S1–S6.

828 [Repository, version and access details of the code and derived data to be added by
829 the authors.]

830 **Acknowledgments**

831 [Funding sources and grant numbers to be added.]

832 **Appendix A. Supplementary data**

833 Supplementary data to this article (Supplementary Notes S1–S6, Tables S1–S7,
834 Figs. S1–S5 and Algorithm S1) are provided in a separate file.

835 **References**

- 836 [1] Berecibar M, Gandiaga I, Villarreal I, Omar N, Van Mierlo J, Van den Bossche P.
837 Critical review of state of health estimation methods of Li-ion batteries for real
838 applications. *Renew Sustain Energy Rev* 2016;56:572–87.
- 839 [2] Xiong R, Li L, Tian J. Towards a smarter battery management system: a critical
840 review on battery state of health monitoring methods. *J Power Sources*
841 2018;405:18–29. <https://doi.org/10.1016/j.jpowsour.2018.10.019>.
- 842 [3] Wang Y, Guo S, Cui Y, et al. A comprehensive review of machine learning-based
843 state of health estimation for lithium-ion batteries: data, features, algorithms, and
844 future challenges. *Renew Sustain Energy Rev* 2025;224:116125.
845 <https://doi.org/10.1016/j.rser.2025.116125>.
- 846 [4] Zhu J, Wang Y, Huang Y, et al. Data-driven capacity estimation of commercial
847 lithium-ion batteries from voltage relaxation. *Nat Commun* 2022;13:2261.
848 <https://doi.org/10.1038/s41467-022-29837-w>.
- 849 [5] Figgenger J, van Ouwerkerk J, Haberschusz D, et al. Multi-year field measurements
850 of home storage systems and their use in capacity estimation. *Nat Energy*
851 2024;9:1438–47. <https://doi.org/10.1038/s41560-024-01620-9>.
- 852 [6] Pastor-Fernández C, Yu TF, Widanage WD, Marco J. Critical review of non-
853 invasive diagnosis techniques for quantification of degradation modes in lithium-
854 ion batteries. *Renew Sustain Energy Rev* 2019;109:138–59.
855 <https://doi.org/10.1016/j.rser.2019.03.060>.

- [7] Bilfinger P, Rosner P, Schreiber M, et al. Battery pack diagnostics for electric vehicles: transfer of differential voltage and incremental capacity analysis from cell to vehicle level. *eTransportation* 2024;22:100356. <https://doi.org/10.1016/j.etrans.2024.100356>.
- [8] Roman D, Saxena S, Robu V, Pecht M, Flynn D. Machine learning pipeline for battery state-of-health estimation. *Nat Mach Intell* 2021;3:447–56. <https://doi.org/10.1038/s42256-021-00312-3>.
- [9] Zhang Y, Tang Q, Zhang Y, Wang J, Stimming U, Lee AA. Identifying degradation patterns of lithium ion batteries from impedance spectroscopy using machine learning. *Nat Commun* 2020;11:1706. <https://doi.org/10.1038/s41467-020-15235-7>.
- [10] Luo Z, Gao Z, Cai Z, et al. A review of on-board electrochemical impedance spectroscopy for low-cost in situ characterization technology. *J Energy Storage* 2026;141:119153. <https://doi.org/10.1016/j.est.2025.119153>.
- [11] Lin Y, Grosu C, Schreiber M, et al. Electrochemical impedance spectroscopy for onboard application: impact of the load conditions on the impedance relaxation behavior. *J Power Sources* 2025;654:237690. <https://doi.org/10.1016/j.jpowsour.2025.237690>.
- [12] Barai A, Uddin K, Widanage WD, McGordon A, Jennings P. A study of the influence of measurement timescale on internal resistance characterisation methodologies for lithium-ion cells. *Sci Rep* 2018;8:21. <https://doi.org/10.1038/s41598-017-18424-5>.
- [13] Li AG, Fahmy YA, Wu MM, et al. Fast time-domain impedance spectroscopy of lithium-ion batteries using pulse perturbation. In: 2022 IEEE Transportation Electrification Conference & Expo (ITEC); 2022, p. 155–60. <https://doi.org/10.1109/ITEC53557.2022.9813868>.
- [14] Birkel CR, Roberts MR, McTurk E, Bruce PG, Howey DA. Degradation diagnostics for lithium ion cells. *J Power Sources* 2017;341:373–86.
- [15] Barai A, Uddin K, Dubarry M, et al. A comparison of methodologies for the non-invasive characterisation of commercial Li-ion cells. *Prog Energy Combust Sci* 2019;72:1–31. <https://doi.org/10.1016/j.pecs.2019.01.001>.
- [16] Li AG, Wang W, West AC, Preindl M. Health and performance diagnostics in Li-ion batteries with pulse-injection-aided machine learning. *Appl Energy* 2022;315:119005. <https://doi.org/10.1016/j.apenergy.2022.119005>.

- [17] Tao S, Ma R, Zhao Z, et al. Generative learning assisted state-of-health estimation for sustainable battery recycling with random retirement conditions. Nat Commun 2024;15:10154. <https://doi.org/10.1038/s41467-024-54454-0>.
- [18] Li M, Xu W, Zhang S, et al. State of health estimation and battery management: a review of health indicators, models and machine learning. Materials 2025;18(1):145. <https://doi.org/10.3390/ma18010145>.
- [19] Kulkarni A, Teodorescu R. Computationally efficient machine-learning-based online battery state of health estimation. arXiv 2024. arXiv:2406.06151.
- [20] Das K, Kumar R. Electric vehicle battery capacity degradation and health estimation using machine-learning techniques: a review. Clean Energy 2023;7(6):1268–81.
- [21] Sylvestrin GR, Maciel JN, Ando Junior OH. Comprehensive feature extraction for battery health prognostics: identifying predictive indicators of state of health. J Power Sources 2026;668:239389. <https://doi.org/10.1016/j.jpowsour.2026.239389>.
- [22] Fu P, Chu L, Hou Z, et al. Transfer learning and vision transformer based state-of-health prediction of lithium-ion batteries. arXiv 2022. arXiv:2209.05253.
- [23] Li Y, Gao G, Chen K, et al. State-of-health prediction of lithium-ion batteries using feature fusion and a hybrid neural network model. Energy 2025;319:135163. <https://doi.org/10.1016/j.energy.2025.135163>.
- [24] Zhao Y, Geng L, Meng W, et al. Cross-attention-based hybrid feature fusion network for state-of-health estimation of lithium-ion batteries. Sci Rep 2026;16:17800. <https://doi.org/10.1038/s41598-026-48317-5>.
- [25] Jin P, Yang S, Xu X, et al. Multiview state-of-health estimation for lithium-ion batteries using time–frequency image fusion and attention-based deep learning. PLoS One 2025;20(11):e0335351. <https://doi.org/10.1371/journal.pone.0335351>.
- [26] Pandit R, Ahlawat N. A standardized comparative framework for machine learning techniques in lithium-ion battery state of health estimation. Future Batteries 2025;7:100099. <https://doi.org/10.1016/j.fub.2025.100099>.
- [27] Bairwa B, Pareek K, Jadoun VK. Cycle based state of health estimation of lithium ion cells using deep learning architectures. Sci Rep 2025;15:37078. <https://doi.org/10.1038/s41598-025-20995-7>.
- [28] Nowacki B, Ramamurthy J, Thelen A, Tischer C, Pint CL, Hu C. Rapid estimation of lithium-ion battery capacity and resistances from short duration

924 current pulses. J Power Sources 2025;628:235813.
 925 <https://doi.org/10.1016/j.jpowsour.2024.235813>.

926 [29] Tao S, Ma R, Chen Y, et al. Rapid and sustainable battery health diagnosis for
 927 recycling pretreatment using fast pulse test and random forest machine learning. J
 928 Power Sources 2024;597:234156.
 929 <https://doi.org/10.1016/j.jpowsour.2024.234156>.

930 [30] Ran A, Cheng M, Chen S, et al. Fast remaining capacity estimation for lithium-
 931 ion batteries based on short-time pulse test and Gaussian process regression.
 932 Energy Environ Mater 2023;6:e12386. <https://doi.org/10.1002/eem2.12386>.

933 [31] Nowacki B, Schmitt T, Aquino P, Hu C. Fine-tuning for rapid capacity
 934 estimation of lithium-ion batteries. Energy Storage Mater 2025;81:104425.
 935 <https://doi.org/10.1016/j.ensm.2025.104425>.

936 [32] Jones PK, Stimming U, Lee AA. Impedance-based forecasting of lithium-ion
 937 battery performance amid uneven usage. Nat Commun 2022;13:4806.
 938 <https://doi.org/10.1038/s41467-022-32422-w>.

939 [33] Lu J, Xiong R, Tian J, Wang C, Sun F. Deep learning to estimate lithium-ion
 940 battery state of health without additional degradation experiments. Nat Commun
 941 2023;14:2760. <https://doi.org/10.1038/s41467-023-38458-w>.

942 [34] Che Y, Lam VN, Rhyu J, et al. Diagnostic-free onboard battery health
 943 assessment. Joule 2025;9:102010. <https://doi.org/10.1016/j.joule.2025.102010>.

944 [35] Sohail M, Tanveer M, Kim HS. Trustworthy machine learning and mathematical
 945 modelling for lithium-ion battery state-of-health estimation. Mathematics
 946 2026;14(11):1879. <https://doi.org/10.3390/math14111879>.

947 [36] Barua N, Collins MD, Islam MS, et al. Predictor structure modulates validation
 948 inflation and cell-transfer reliability in battery state-of-health estimation. Batteries
 949 2026;12(9):342. <https://doi.org/10.3390/batteries12090342>.

950 [37] Zhang H, Gui X, Zheng S, et al. BatteryML: an open-source platform for
 951 machine learning on battery degradation. In: International Conference on
 952 Learning Representations (ICLR); 2024.

953 [38] Bai S, Kolter JZ, Koltun V. An empirical evaluation of generic convolutional and
 954 recurrent networks for sequence modeling. arXiv 2018. arXiv:1803.01271.

955 [39] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. In: Advances
 956 in Neural Information Processing Systems 30; 2017, p. 5998–6008.

957 [40] Nie Y, Nguyen NH, Sinthong P, Kalagnanam J. A time series is worth 64 words:
958 long-term forecasting with transformers. In: International Conference on
959 Learning Representations (ICLR); 2023. [https://openreview.net/forum?](https://openreview.net/forum?id=Jbdc0vTOcol)
960 [id=Jbdc0vTOcol](https://openreview.net/forum?id=Jbdc0vTOcol).

961 [41] Gorishniy Y, Rubachev I, Khrulkov V, Babenko A. Revisiting deep learning
962 models for tabular data. In: Advances in Neural Information Processing Systems
963 34; 2021, p. 18932–43.

964 [42] Liu H, Li C, Hu X, et al. Multi-modal framework for battery state of health
965 evaluation using open-source electric vehicle data. *Nat Commun* 2025;16:1137.
966 <https://doi.org/10.1038/s41467-025-56485-7>.

967 [43] Huang X, Tao S, Liang C, et al. iMOE: prediction of second-life battery
968 degradation trajectory using interpretable mixture of experts. *Nat Commun*
969 2026;17:2549. <https://doi.org/10.1038/s41467-026-69369-1>.

970 [44] Lu Z, Mo H, Wang Y. MSSformer: a model for accurate and stable state of
971 health estimation of lithium-ion batteries. *Electr Power Syst Res*
972 2026;254:112661. <https://doi.org/10.1016/j.epsr.2025.112661>.

973 [45] Tang P, Qiu Z, Yao Z, et al. Lithium-ion battery RUL prediction based on
974 optimized VMD-SSA-PatchTST algorithm. *Sci Rep* 2025;15:26824.
975 <https://doi.org/10.1038/s41598-025-11934-7>.

976 [46] Labib KM, Rasool I, Ahmed S. Operator-theoretic and physics-guided sequence
977 modeling of lithium-ion battery voltage dynamics. *arXiv* 2026.
978 [arXiv:2605.01693](https://arxiv.org/abs/2605.01693).

979 [47] Wang F, Zhai Z, Zhao Z, Di Y, Chen X. Physics-informed neural network for
980 lithium-ion battery degradation stable modeling and prognosis. *Nat Commun*
981 2024;15:4332. <https://doi.org/10.1038/s41467-024-48779-z>.

982 [48] Che Y, Zheng Y, Rhyu J, et al. Mechanistically guided residual learning for
983 battery state monitoring throughout life. *Nat Commun* 2026;17:855.
984 <https://doi.org/10.1038/s41467-025-67565-z>.

985 [49] Guo W, Vilsen SB, Li Y, Verma A, Stroe DI, Brandell D. Uncovering the impact
986 of battery design parameters on health and lifetime using short charging
987 segments. *Energy Environ Sci* 2025;18:8462–74.
988 <https://doi.org/10.1039/D5EE03268G>.

989 [50] Sundararajan M, Taly A, Yan Q. Axiomatic attribution for deep networks. In:
990 Proceedings of the 34th International Conference on Machine Learning (ICML),
991 PMLR 70; 2017, p. 3319–28.

992 **CRedit authorship contribution statement**

993 [Author 1]: Conceptualization, Methodology, Software, Formal analysis, Writing
994 – original draft. [Author 2]: Validation, Visualization, Writing – review & editing.
995 [Corresponding Author]: Supervision, Funding acquisition, Writing – review &
996 editing.